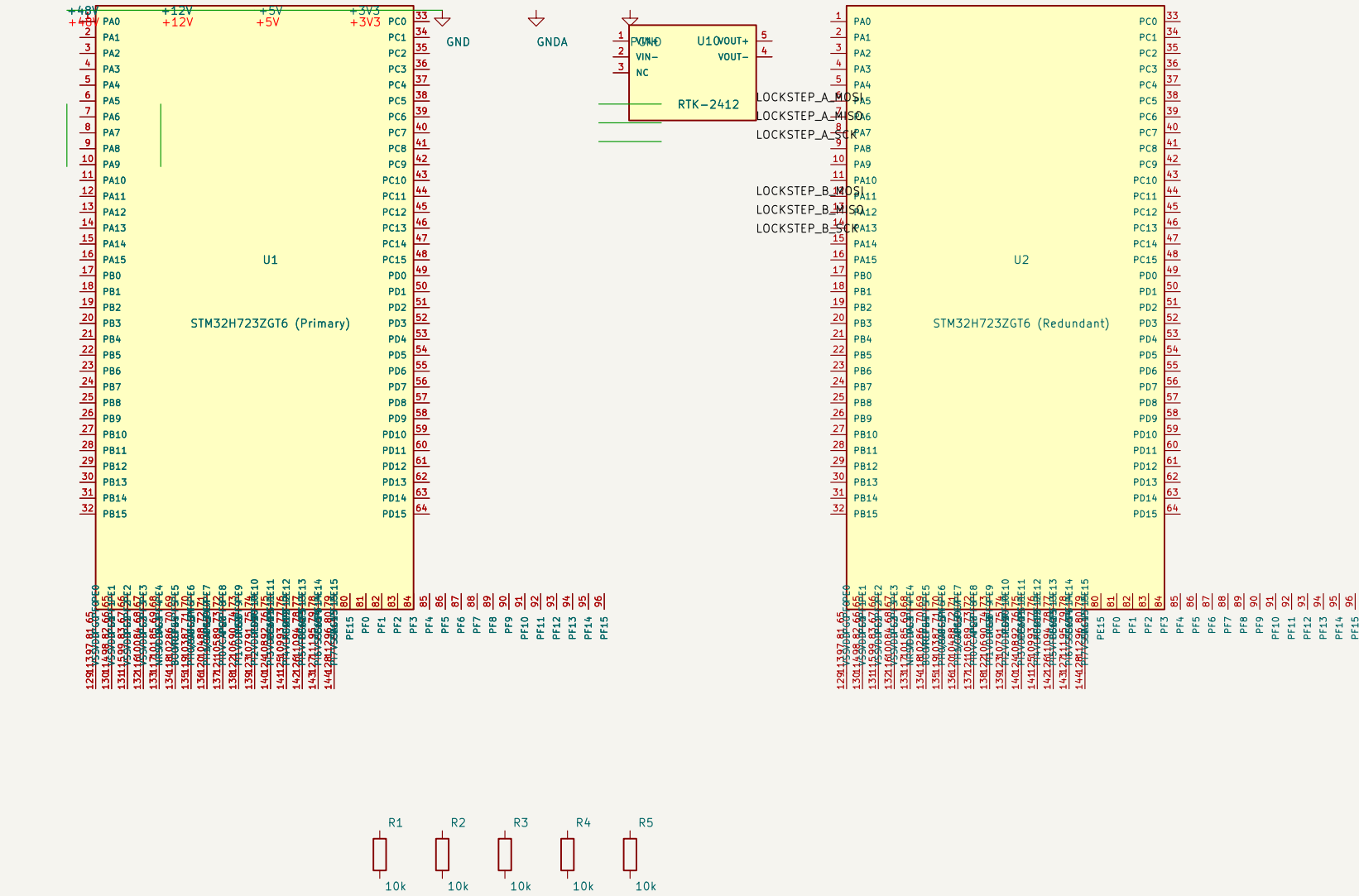


PAL POD HALBACH LEVITATION CONTROLLER – REV A0 – PLACEHOLDER

SAFETY–CRITICAL – Full design review + FMEA required before first power–on
Reference: hardware/electrical/block–diagrams/levitation–controller.md

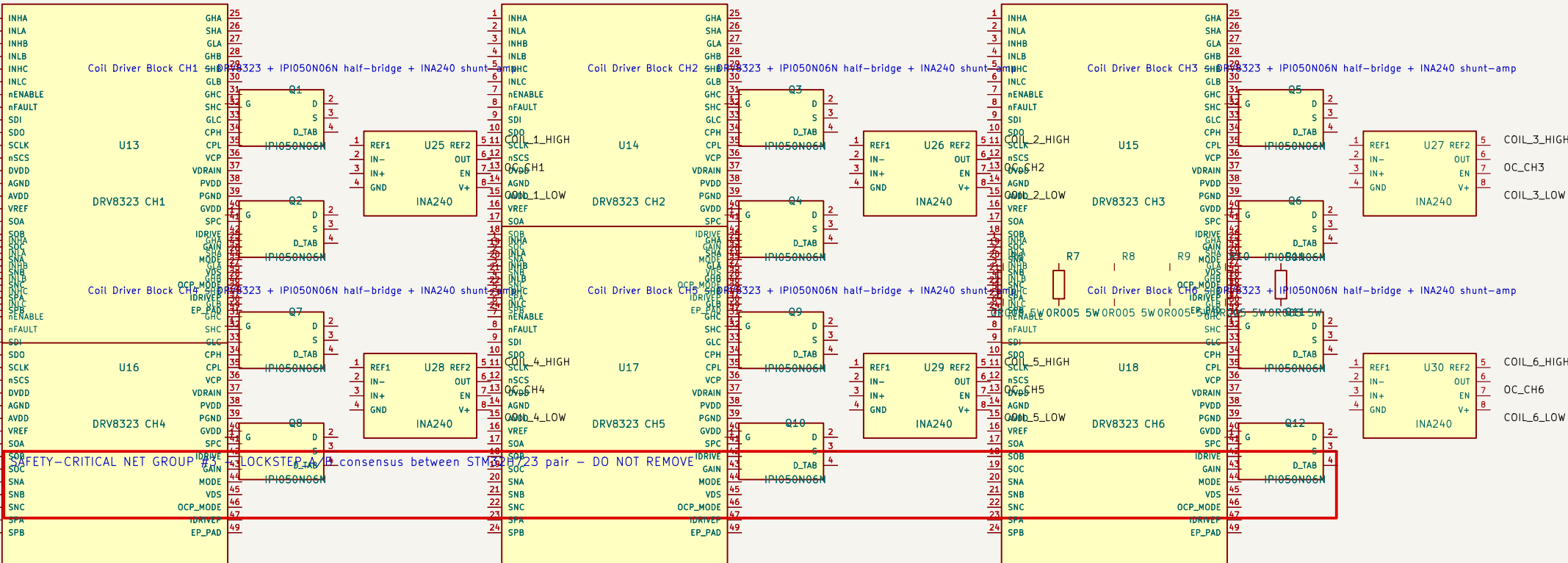
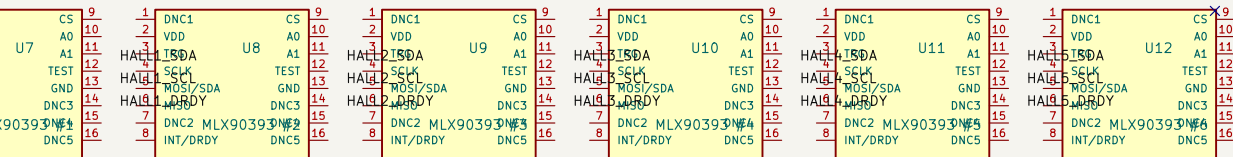
SAFETY–CRITICAL NET GROUP #1 – OC_LATCH / nFAULT / WDOG – DO NOT REMOVE

Free: 12V–10V–5V–3V3–GND
#PWR001: 12V–10V–5V–3V3–GND
#PWR002: 12V–10V–5V–3V3–GND
#PWR003: 12V–10V–5V–3V3–GND
#PWR004: 12V–10V–5V–3V3–GND
#PWR005: 12V–10V–5V–3V3–GND
#PWR006: 12V–10V–5V–3V3–GND
Cross–strapped SPI: LOCKSTEP_A (U1 SPI2–>U2 SPI2), LOCKSTEP_B (U2 SPI3–>U1 SPI3)
#PWR007: 12V–10V–5V–3V3–GND
For 48V: cascade with LM5155 boost per driver–stage spec

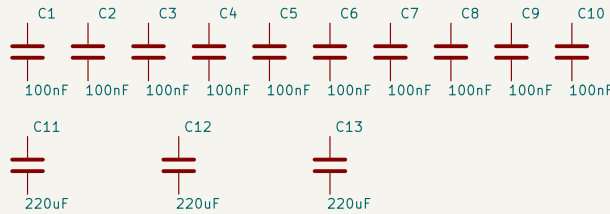


SAFETY–CRITICAL NET GROUP #2 – ESTOP_IN / ESTOP_LATCH – hardwired to MAX706 MR – DO NOT REMOVE

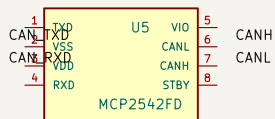
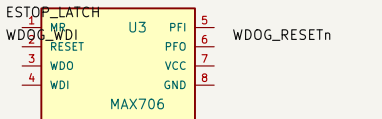
6x MLX90393 3D hall sensors – 3 around column top + 3 around expected orb position



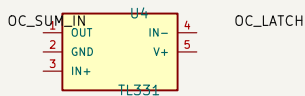
E–STOP INPUT (rear panel) – hardwired to MAX706 MR pin, coils de–energize <10ms



U3: MAX706 window WDT + MR (manual reset from E–stop) + PFI (power–fail input)



U4: TL331 – hardware overcurrent latch, independent of MCU firmware.



U6: Aux MCU – coil temperature monitoring + heatsink fan PWM

